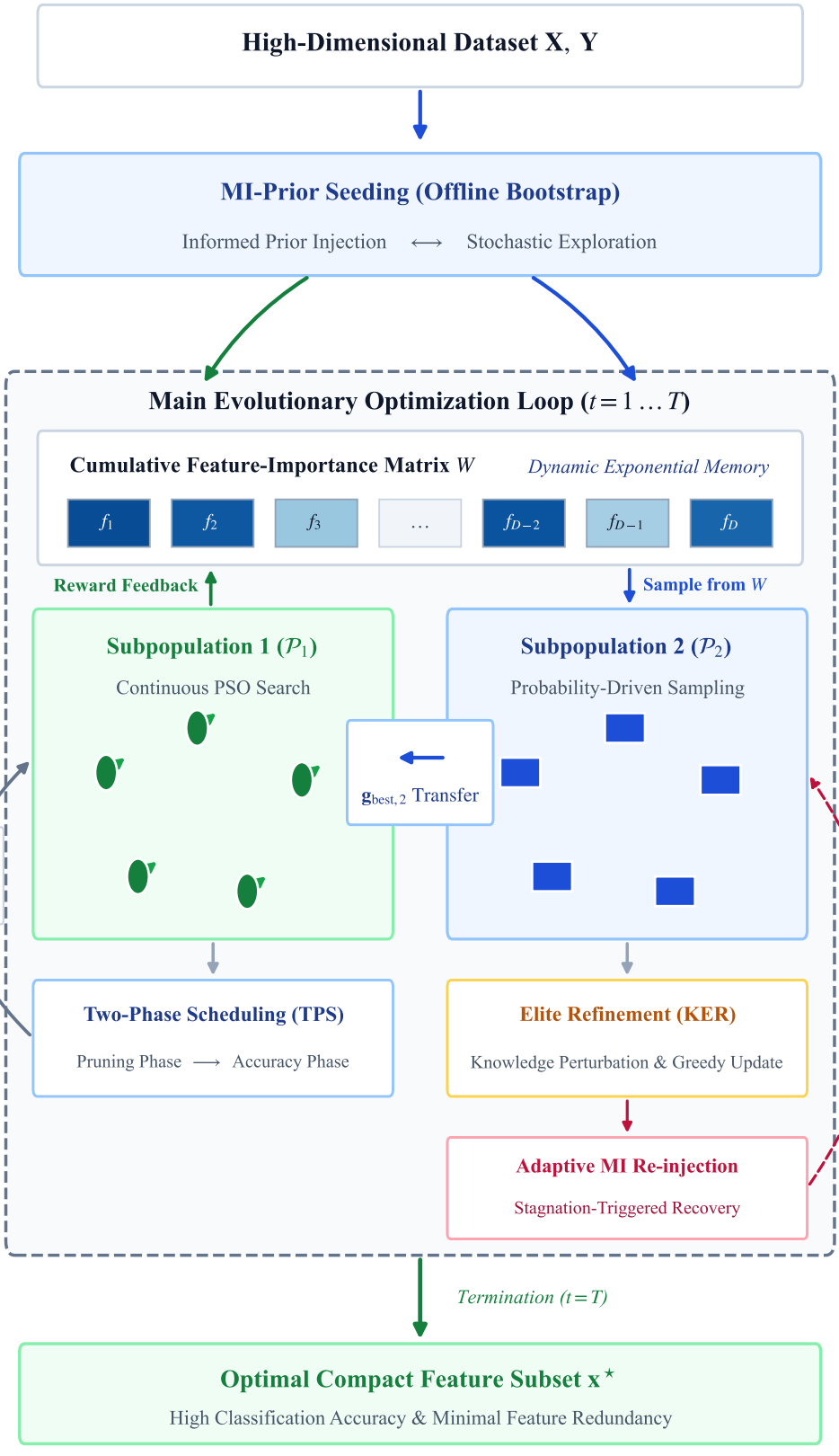

High-Dimensional Dataset X, Y
MI-Prior Seeding (Offline Bootstrap)
Informed Prior Injection ⟷ Stochastic Exploration
Main Evolutionary Optimization Loop (t = 1 … T)
Cumulative Feature-Importance Matrix W
Dynamic Exponential Memory
f1 f2 f3 … fD−2 fD−1 fD
Reward Feedback
Sample from W
Subpopulation 1 (P1)
Continuous PSO Search
Subpopulation 2 (P2)
Probability-Driven Sampling
gbest, 2 Transfer
t ← t + 1
Two-Phase Scheduling (TPS)
Pruning Phase ⟶ Accuracy Phase
Elite Refinement (KER)
Knowledge Perturbation & Greedy Update
Adaptive MI Re-injection
Stagnation-Triggered Recovery
Termination (t = T)
Optimal Compact Feature Subset x⋆
High Classification Accuracy & Minimal Feature Redundancy